

Project Initialization and Planning Phase

Date	08 October 2026
Team ID	UNM1611BHAVANA
Project Title	AI FitTrack API - AI-Powered Fitness Tracking REST API
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform personal fitness tracking using a secure REST API enhanced with AI-driven recommendations, boosting user engagement and health outcomes. It tackles scattered manual tracking and the lack of personalised guidance. Key features include workout and nutrition tracking, goal management and AI-based workout and diet recommendations.

Project Overview	
Objective	The primary objective is to build a secure REST API that lets users log workouts, meals and body metrics, set fitness goals, and receive AI-driven workout and diet recommendations and progress insights.
Scope	The project covers a backend API with user authentication, profile management, workout and nutrition tracking, goal management, AI recommendations and progress reports, usable from Postman or any future web/mobile client.
Problem Statement	
Description	Many people track fitness manually or across disconnected apps, with no personalised guidance. Data is scattered and inconsistent and is rarely turned into useful advice.
Impact	A single centralised API gives users consistent records, personalised plans and clear progress feedback, improving motivation, adherence and overall health outcomes.
Proposed Solution	
Approach	A layered Node.js/Express backend (routes, controllers, models) with MongoDB storage, JWT authentication and an AI recommendation module that analyses a user's profile, goals and activity history to suggest workouts and diet plans.
Key Features	- Secure registration and login using JWT - Workout, nutrition and body-metric tracking (CRUD) - Goal setting and progress tracking - AI-based workout and diet recommendations - Progress analytics and summaries - Input validation and centralised error handling

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop/desktop, 4-core CPU (no GPU required)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Frameworks	Backend frameworks	Node.js, Express.js
Libraries	Additional libraries	Mongoose, jsonwebtoken, bcrypt, dotenv, cors, express-validator, Jest, Supertest
Development Environment	IDE / tools	Visual Studio Code, Postman, Git/GitHub
Data		
Data	Source, size, format	User-generated data (profiles, workouts, meals, goals) stored as JSON documents in MongoDB